Reformatted Train Withdrawal Scenario proposed for KMRCL vide letter no BEML/RS(3R)/TC/Withdrawal Scenario/718 dtd: 02.04.2022

(Assumption: Failure scenario as observed at Motorman Level)

Sl	Failure Scenario	Condition calling train withdrawal
1	**Loss of tractive effort**	Steps 1.1 to 1.4 to be checked before witdrawal
1.1	EB Releasing when powering is intended	Revenue can be continued
1.2	HSCB Tripped and reset not happening	Withdraw by isolating faulty VVVF. (Rescue if all 4 HSCB are tripped)
1.3	Traction possible in SCS Cut-Off	Withdraw. (Rescue if traction not possible in SCS cut-off)
1.4	EB is released in DSD Bypass mode	Withdraw (Train to be rescued if EB is not released in SCS Cut-Off, DPL Bypass and DSD Bypass Modes)
2	Loss of braking effort / Failure of brake control circuits in either cab	Wihdraw
3	Loss of MR / BP /PB pressure (below respective	Withdraw
4	Failure of train door control circuits (Entire train) at either cab	Withdraw
5	Failiure of either of the cab doors to open or close (Either physically or loss of electrical signal)	Withdraw
6	Failure of either of the partition doors to open or close	Withdraw only if the door cannot be opened from either side.
7	Failure of consecutive 2 or more saloon doors on the same side of a car even if the failed door can be isolated	Withdraw
8	PWM Generator Failure (In any DMC)	Withdraw
9	Breakage of even one Windshield Glass.	Withdraw
10	Breakage of one or more Window / Door Glass.	Withdraw
11	Failure of structural integrity of any car	Withdraw / Rescue as per merit
12	Failure of driver's cab control which affects revenue operation	Withdraw
13	Degradation of driver's environment in rainy seasor	Withdraw
14	Failure of Gangway / Separation of Gangway	Withdraw
15	Battery under voltage during service	If Battery voltage is shown around 110V in APS page of TCMS and there is no BCG fault, then it is not required to withdraw (DI issue). Otherwise withdraw.
16	Failure of Saloon VAC	Folowing conditions to be checked.
16.1	VAC Running in Emergency Mode	Withdraw if any 2 or more adjacent VACs are running in emergency mode.
16.2	VAC not Running in any mode	Withdraw if any 2 or more adjacent VACs are not running in any mode.
17	Failure of Cab VAC in any of the Cabs	Check for following condition
17.1	Bleeder Fan is working	Withdraw if bleeder fan is not working in the above condition
18	Failiure of Public Announcement (either PRM or Journey Announcement) during service.	There are 8 speakers per car with redundancy. Train should be withdrawan from revenue operation if all 8 speakers of any one car (or more) stop functioning.
19	Passenger is not able to talk to train operator during service	Train can be withdrawan if any 2 consecutive PEAU fail to communicate with TO
20	TO is not able to make manual Announcement during service.	Train can be withdrawan if TO can not address to the passenger either from HMI/Microphone or from ACP
21	Failiure of Public Broadcsting (in TFT) during service.	There are 6 TFT per car. Train should be withdrawan from revenue operation if all 6 TFTs of any car stop functioning.
22	Failure of CCTV HMI	Even if the CCTV HMI fails, data will be recorded in DVR. No need to withdraw.

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Sl	Failure Scenario	Condition calling train withdrawal
23	Poor Ride Quality due to loss of integrity of bogie / structure / wheelset / suspension / drive gear / bearings etc.	Withdraw
24	Failiure of head light high beam in leading direction.	Withdraw if both the head light assembly fail in the leading DMC
25	TCMS CCU is offline or transmision error	If one CCU fails,the other CCU will continue to monitor/control the signals and service operation can be continued.
26	TCMS interface unit (RIO) is offline or transmision error	If one RIO fails,the other RIO will continue to monitor the signals and service operation can be continued.
27	TCMS Control Node (CN) is offline or transmision error	If one CN fails,the other CN will continue to communicate with other sub systems and service operation can be continued.
28	TCMS Video Display Unit Failiure and doesnot reset after switching Off and On the VDU Power Supply CB.	Withdraw
29	APS Major Fault in TCMS	Train to be withdrawn if fault cannot be reset.
30	Speed Restriction is applied on account of BIC cut-	Withdraw, if BIC can not be made normal
31	Speed Restriction is applied on account of Air Spring	Withdraw, if restricted mode does not reset even after Aux-Off and Aux-On the train at buffer end.